Pet-Tracker

MSc Electronic Engineering

Students:
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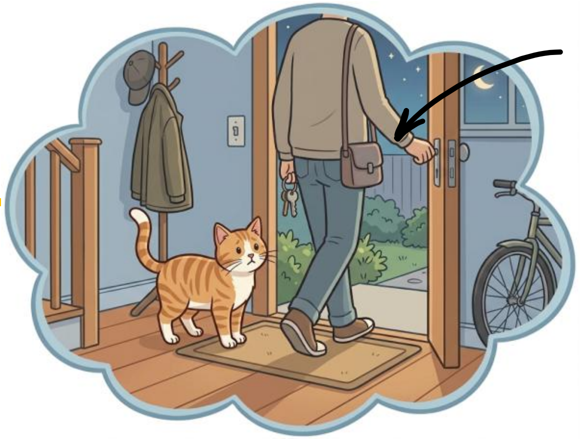
Problem Overview

rest in specific spots



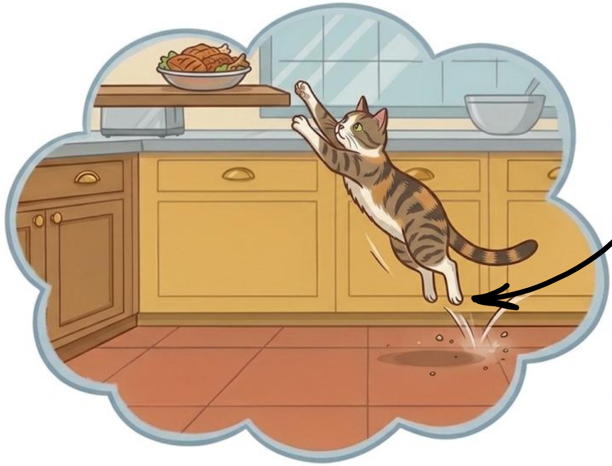
No human supervision

hygiene issues due to shedding

knocking over valuable items

Without human supervision, pets alternate rest and active periods. An untrained pet in restricted areas can cause damage and put itself at risk.

jump onto desks with electronic devices or important paper documents

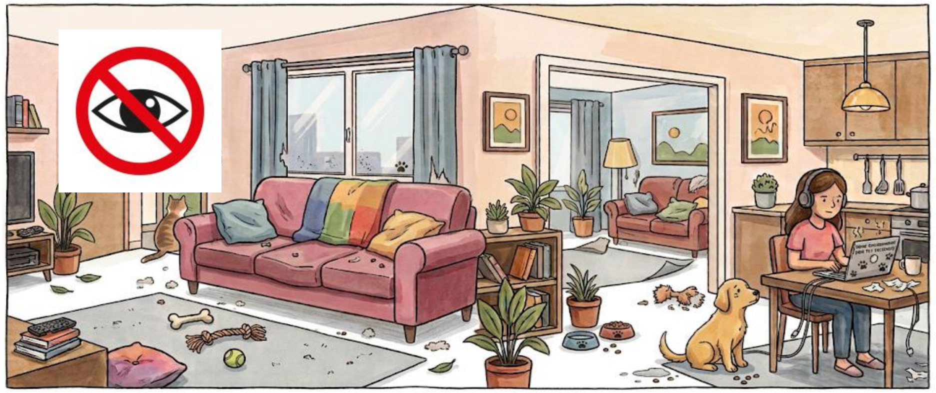


reach exposed food or get too close to appliances

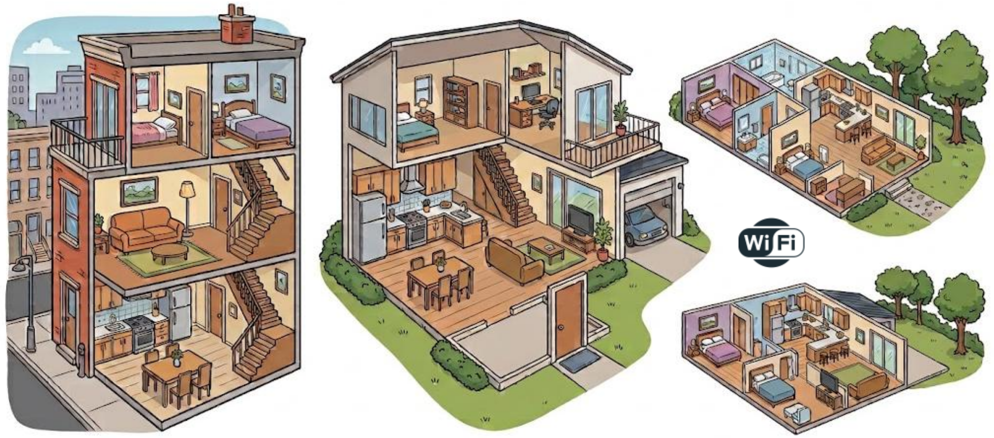
Reference Scenario

WHO? Pet owners unable to constantly monitor their pet.

WHEN? During periods of non-supervision.



WHERE? Indoor environments of any size or layout.



WHAT? Training purpose & Pet localization

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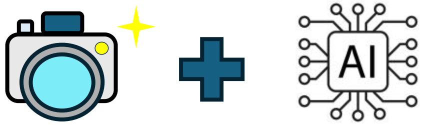
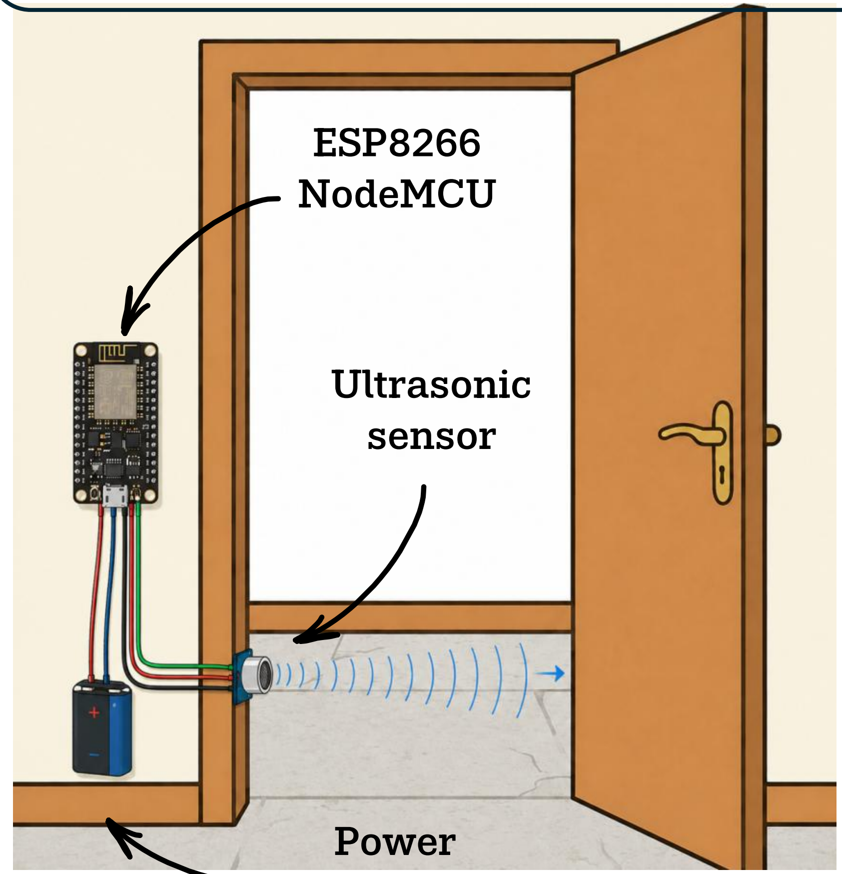
Room transition detection

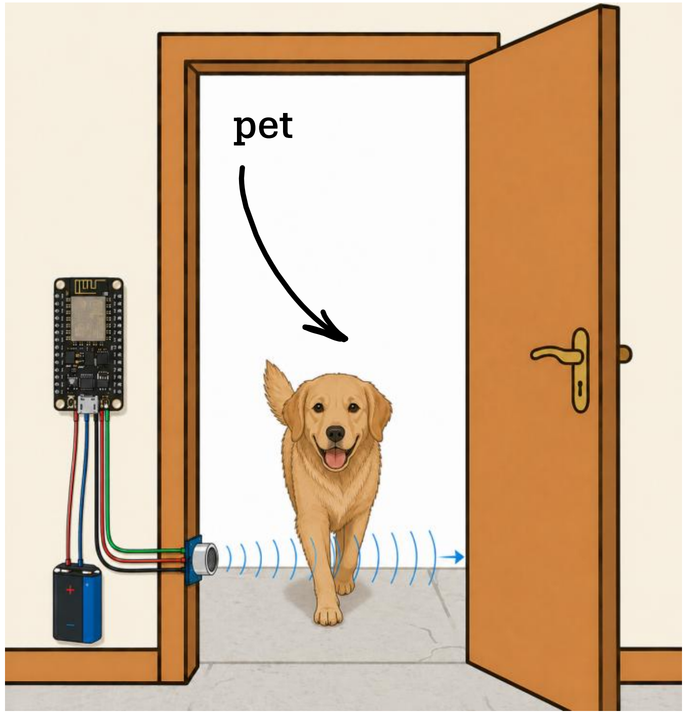
The door becomes a smart object. Each device have to be put in each door to detect **room-to-room transitions**.

After these

HOW can the system distinguish a pet from any other entity?







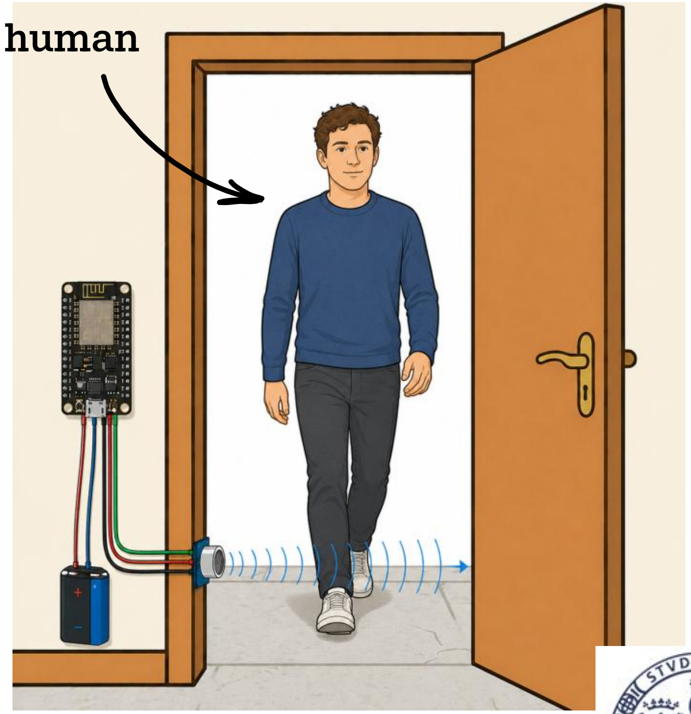
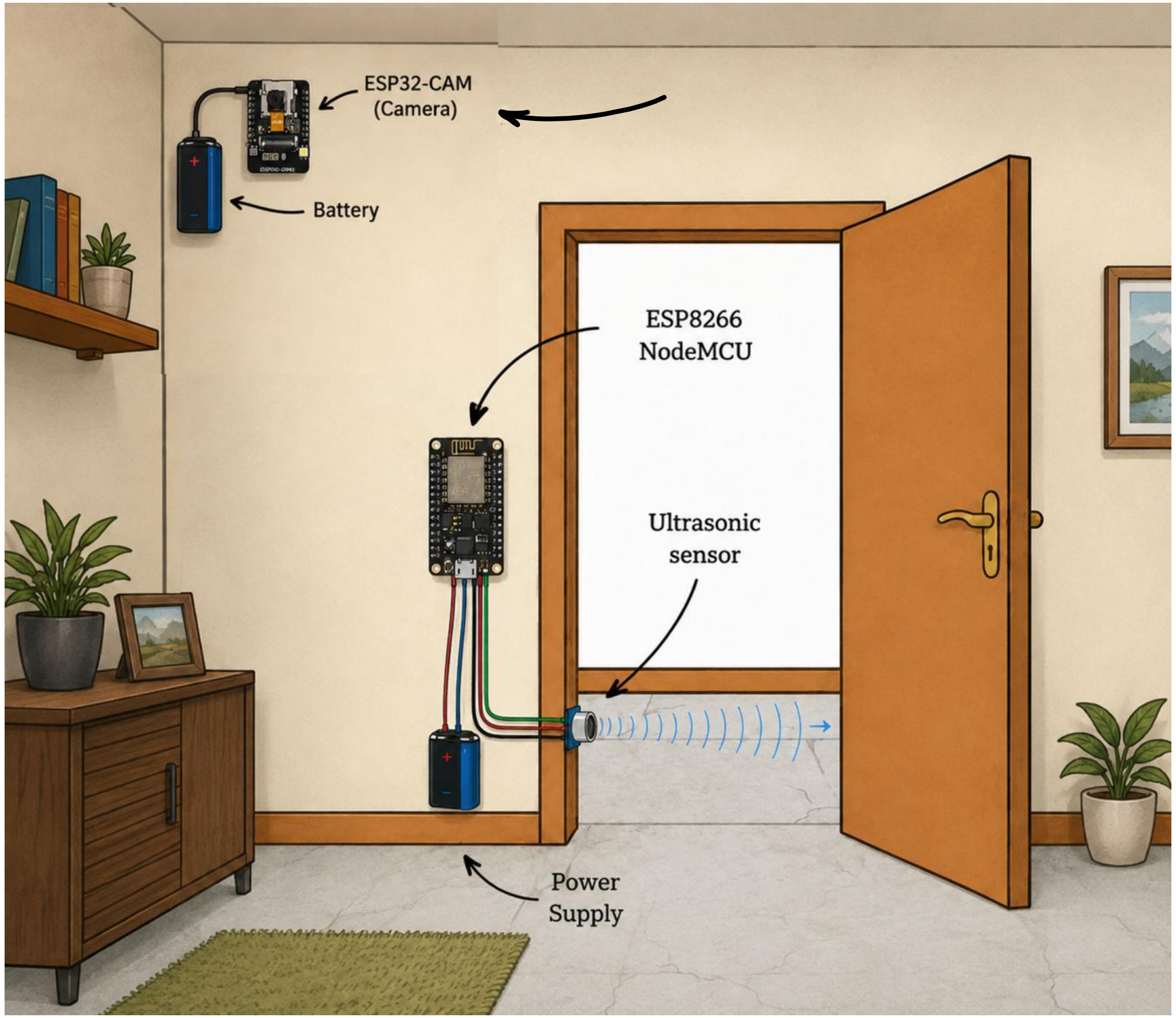
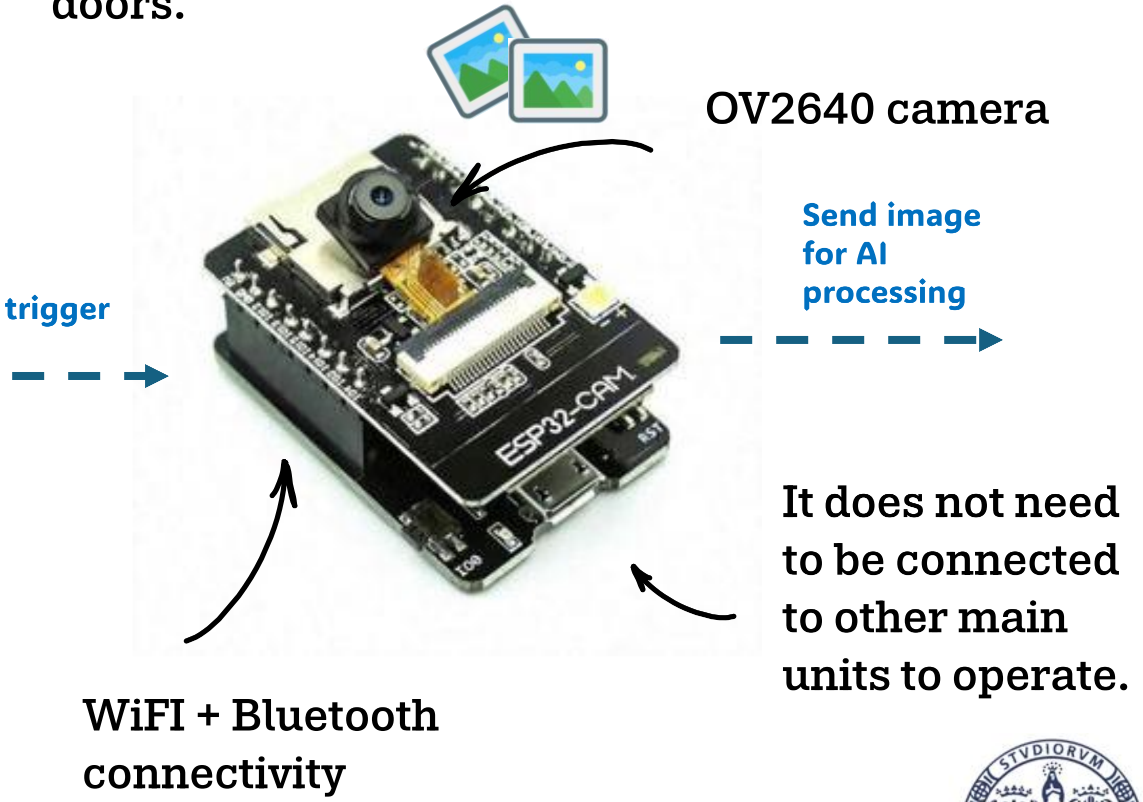


Image acquisition



The room becomes a smart object. Each ESP32-CAM has to be placed in each room to acquire a single image when triggered by the node upon crossing one of the room's entry doors.



Pet Training System

The pet becomes a smart object by attaching the device to either a collar or a harness.

- **Automatic Trigger**: The collar buzzer activates upon the pet entering a restricted room and deactivates upon exit.

System

- **Manual Override**: The administrator can manually turn the buzzer on or off in any room at any time.

Admin

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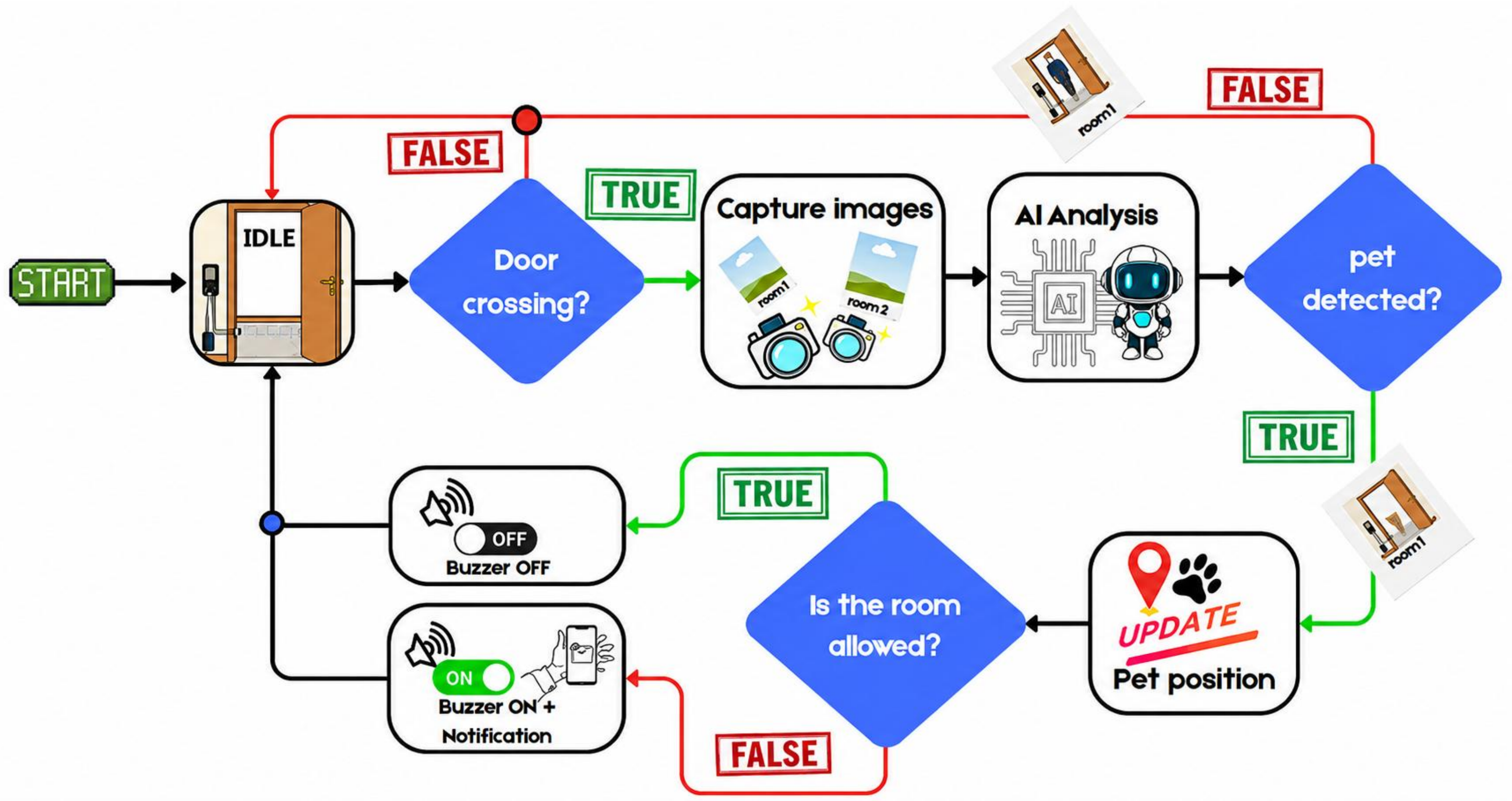
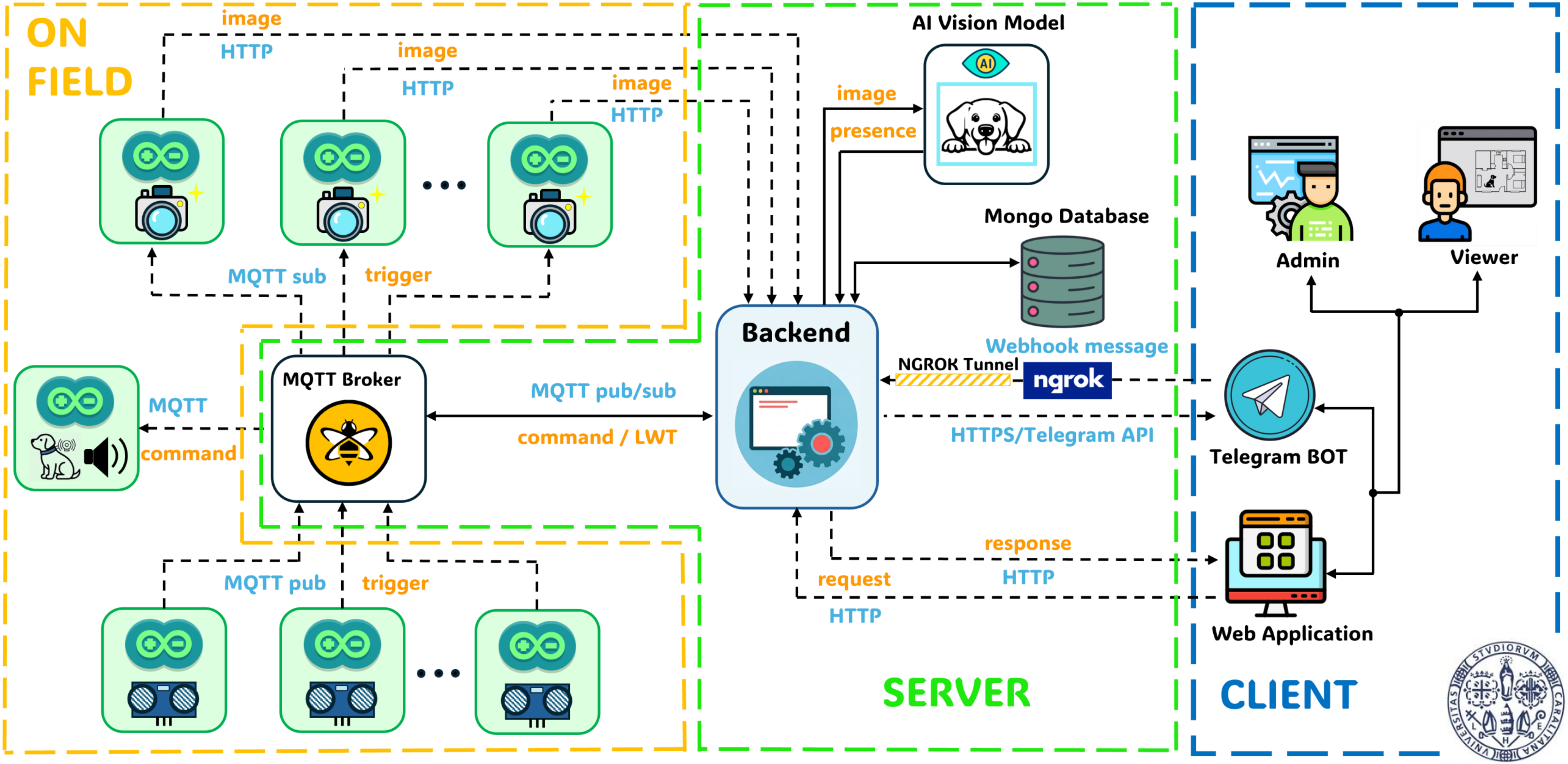
Workflow of the System



Architecture Overview



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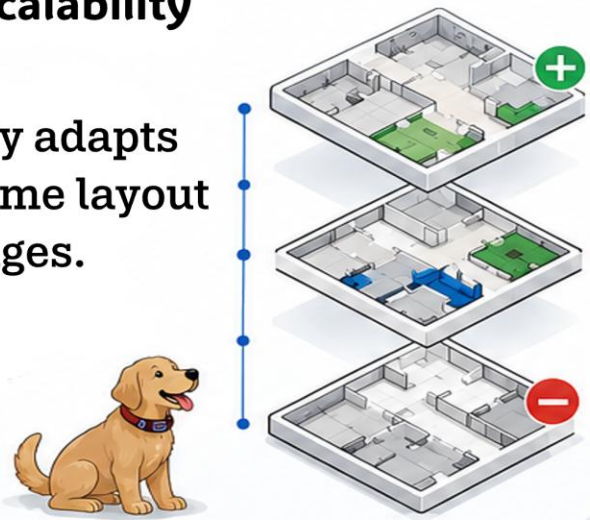
Strenght Points of our System

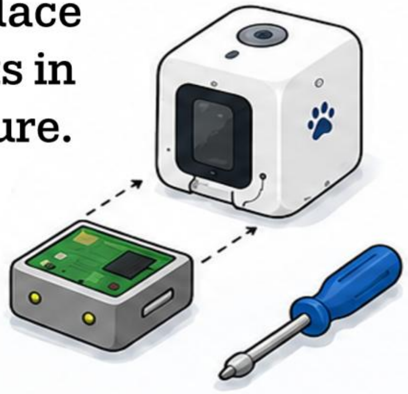
1 Scalability

Easily adapts to home layout changes.



2 Component Replaceability

Easy to replace components in case of failure.



3 System Robustness

Ensures operational continuity even in case of partial failures.

4 Low Latency

Real time user notifications and buzzer activation within 3 s.

5 Privacy

Restricts access to sensitive data.

6 User-Friendly Interface

Intuitive UI suitable for everyone.

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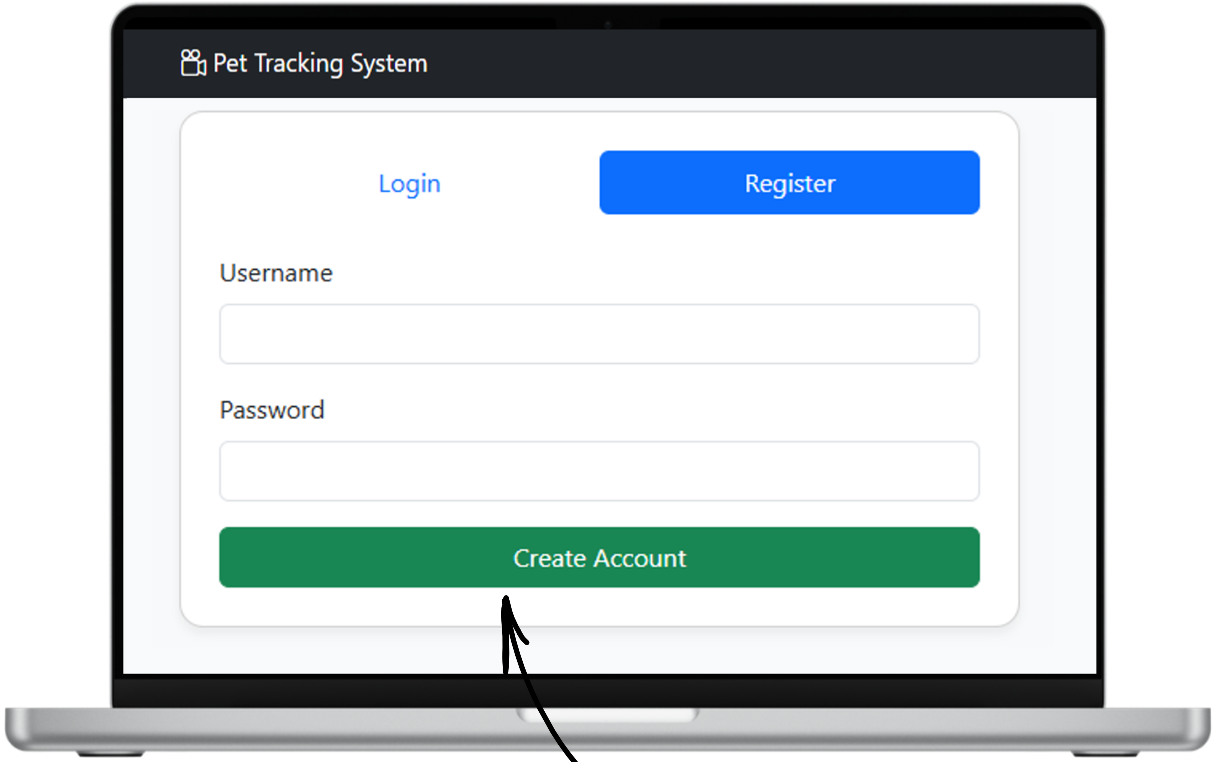
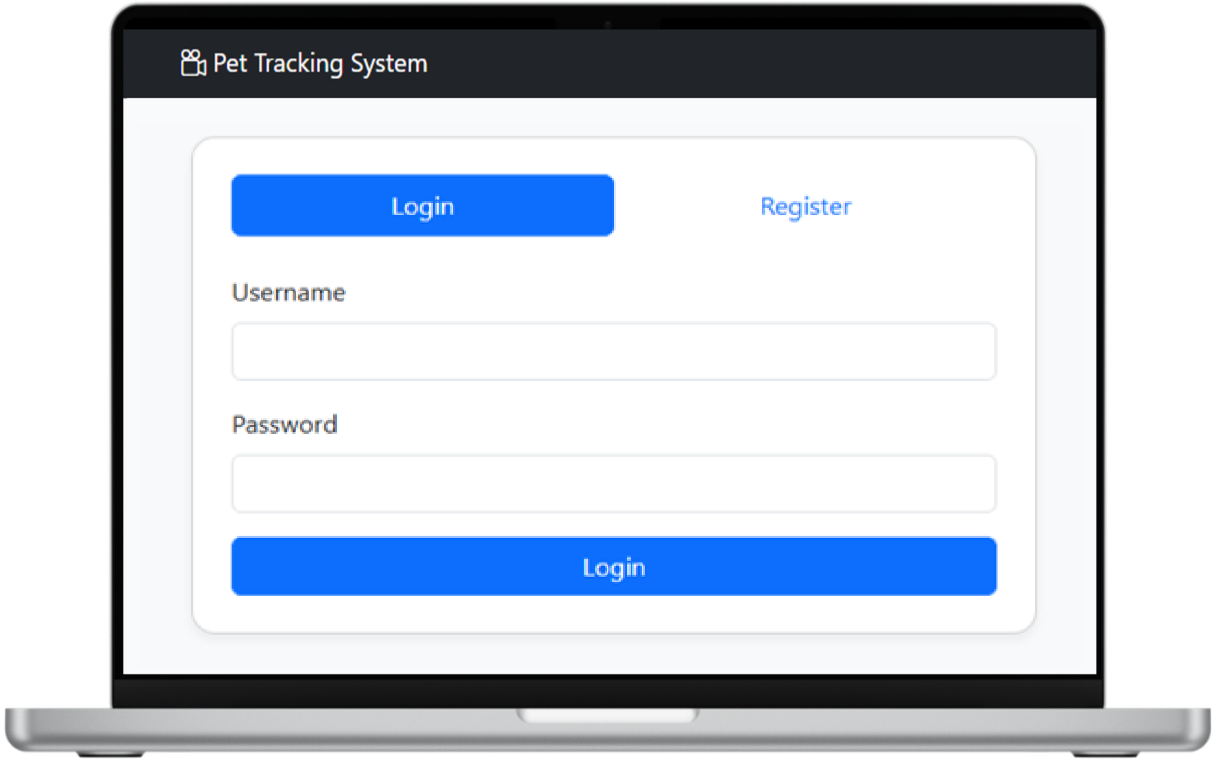
Registration & Login

1° step - Registration



Create Account (e.g.
Username: marco138
Password: IoTlexa138!

2° step - Login



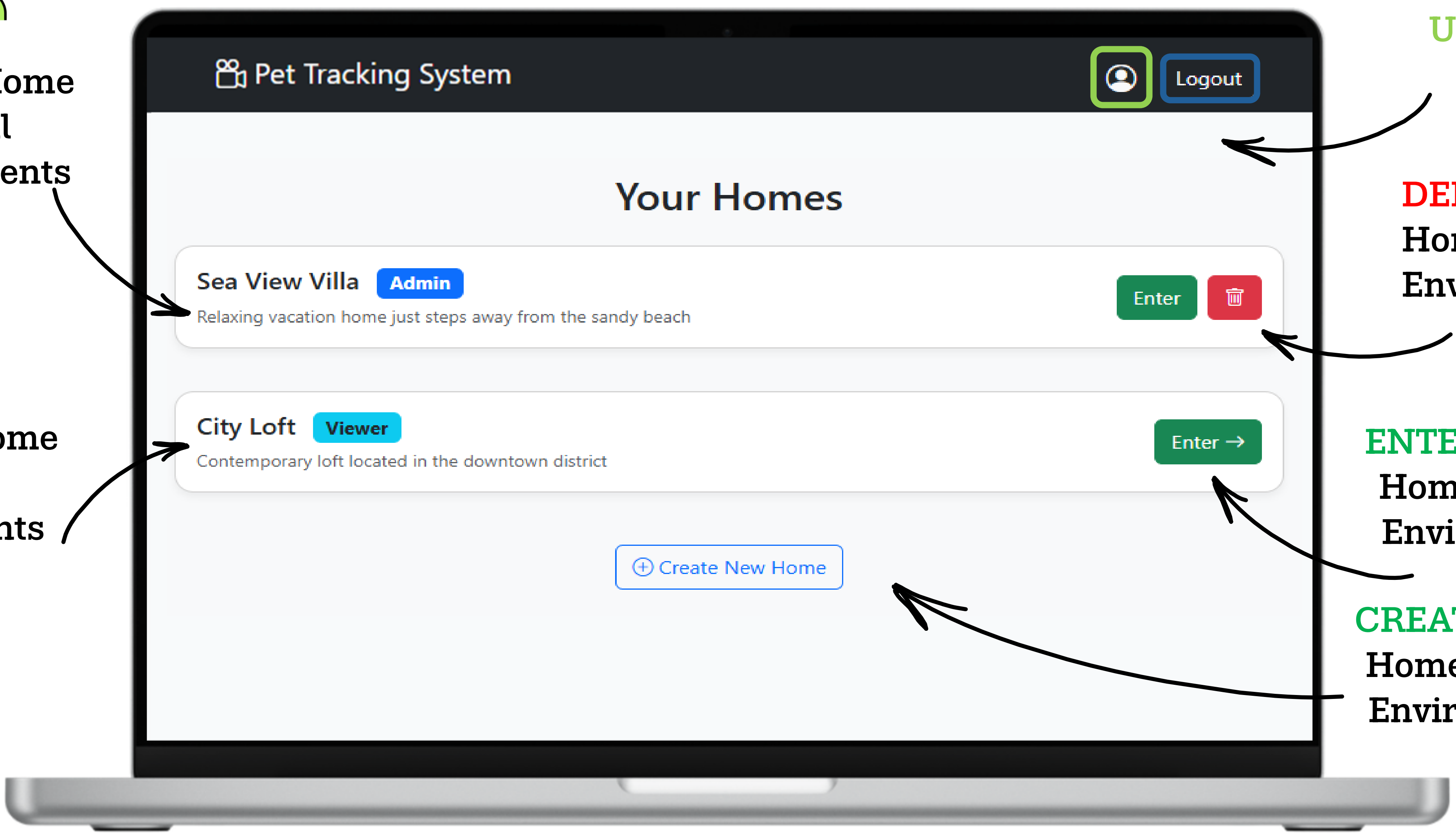
Your Homes

Admin

Created Home Virtual Environments

Viewer

Viewable Home Virtual Environments



Logout & User profile buttons

DELETE the Home Virtual Environment

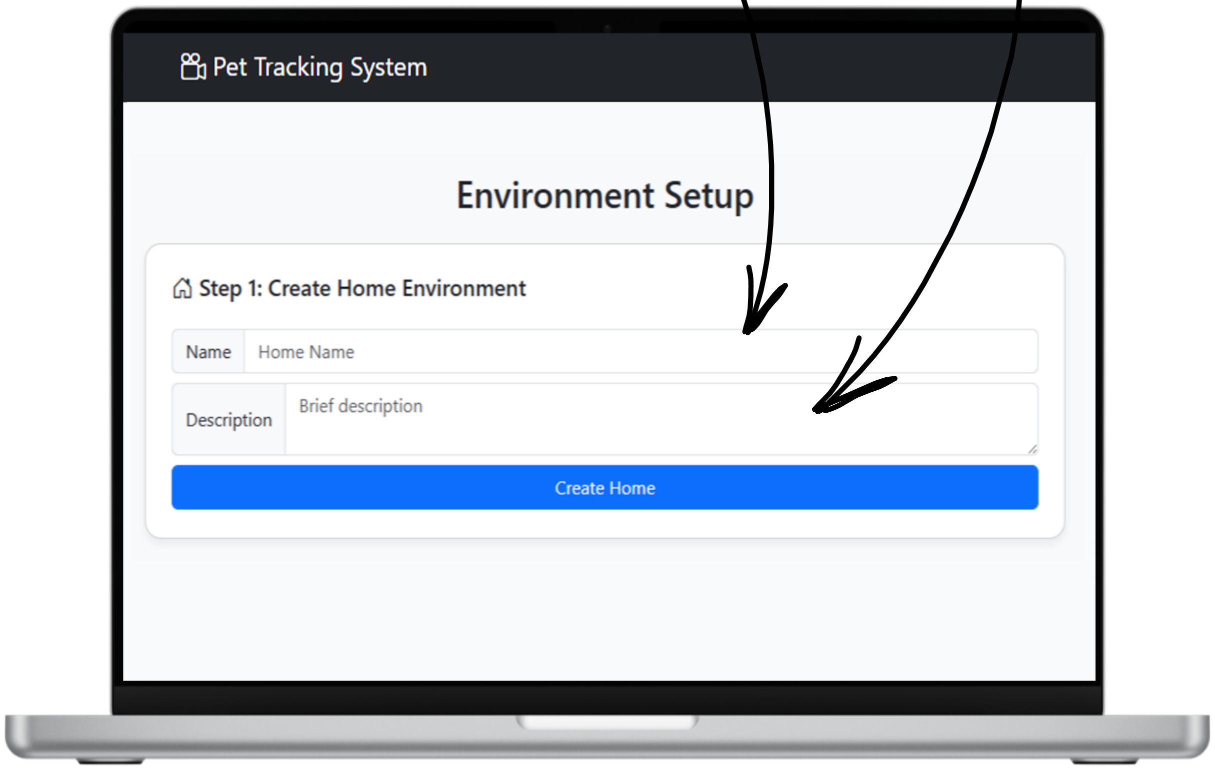
ENTER into the Home Virtual Environment

CREATE a new Home Virtual Environment

Home Environment Creation

Name
(e.g. Marco’s Home)

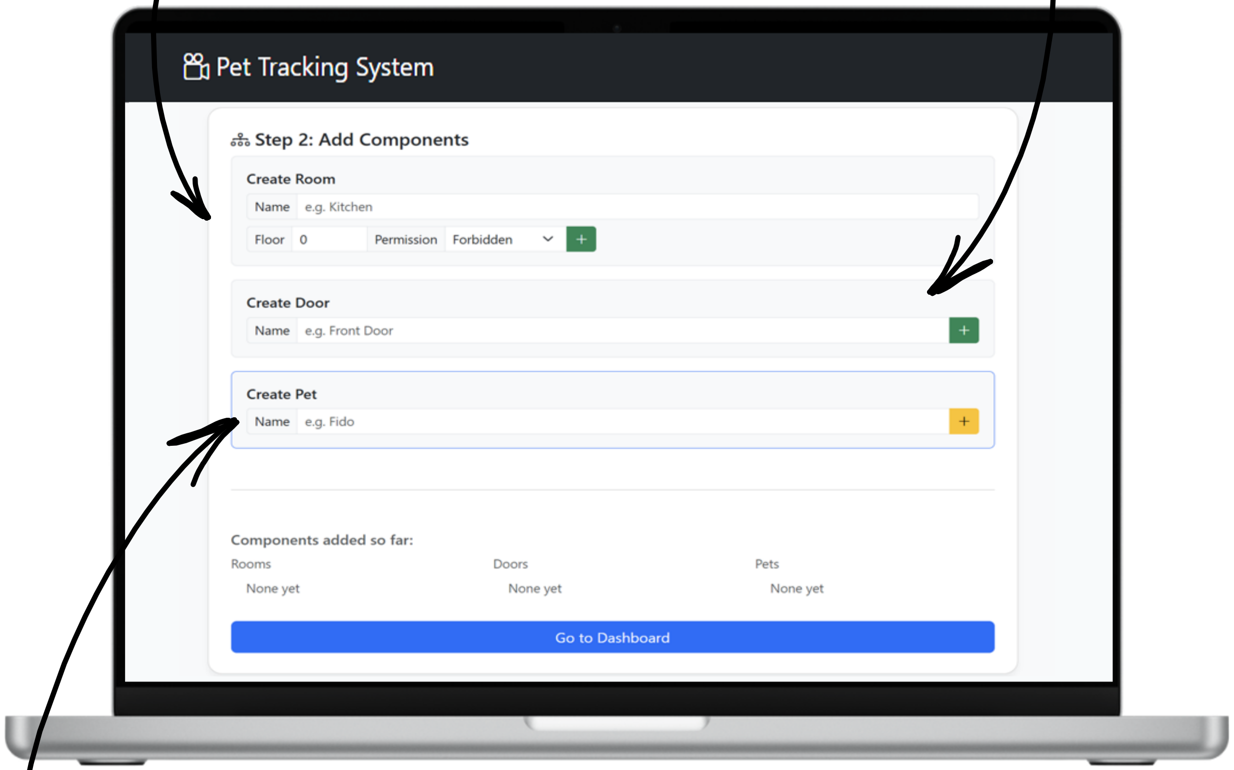
Description
(e.g. Smart Family Home)



1° step - **Create the Home**

Create Room
(e.g. name: Bathroom,
Floor:1,
Permission: Forbidden)

Create Door
(e.g. name:door_u1)



Create Pet
(e.g. name: Fido)

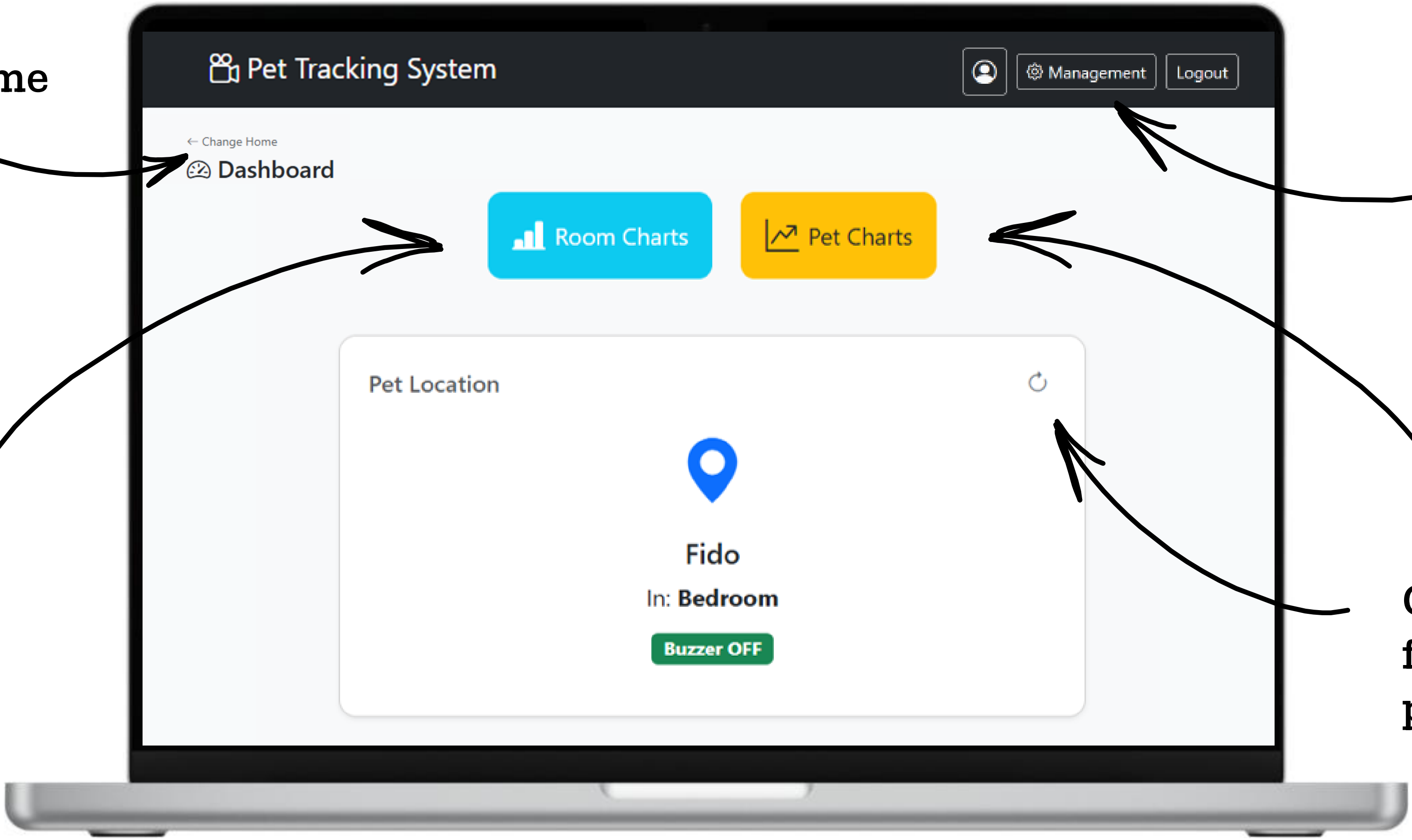
2° step - **Add Components**

Dashboard

After clicking the Enter → button in *Your Homes*

Change Home

Button visible only to the administrator (used to update rooms, doors, pet configuration and to add/remove viewers)



Daily Room Statistics

Statistics on pet violations for the most recent 30 days

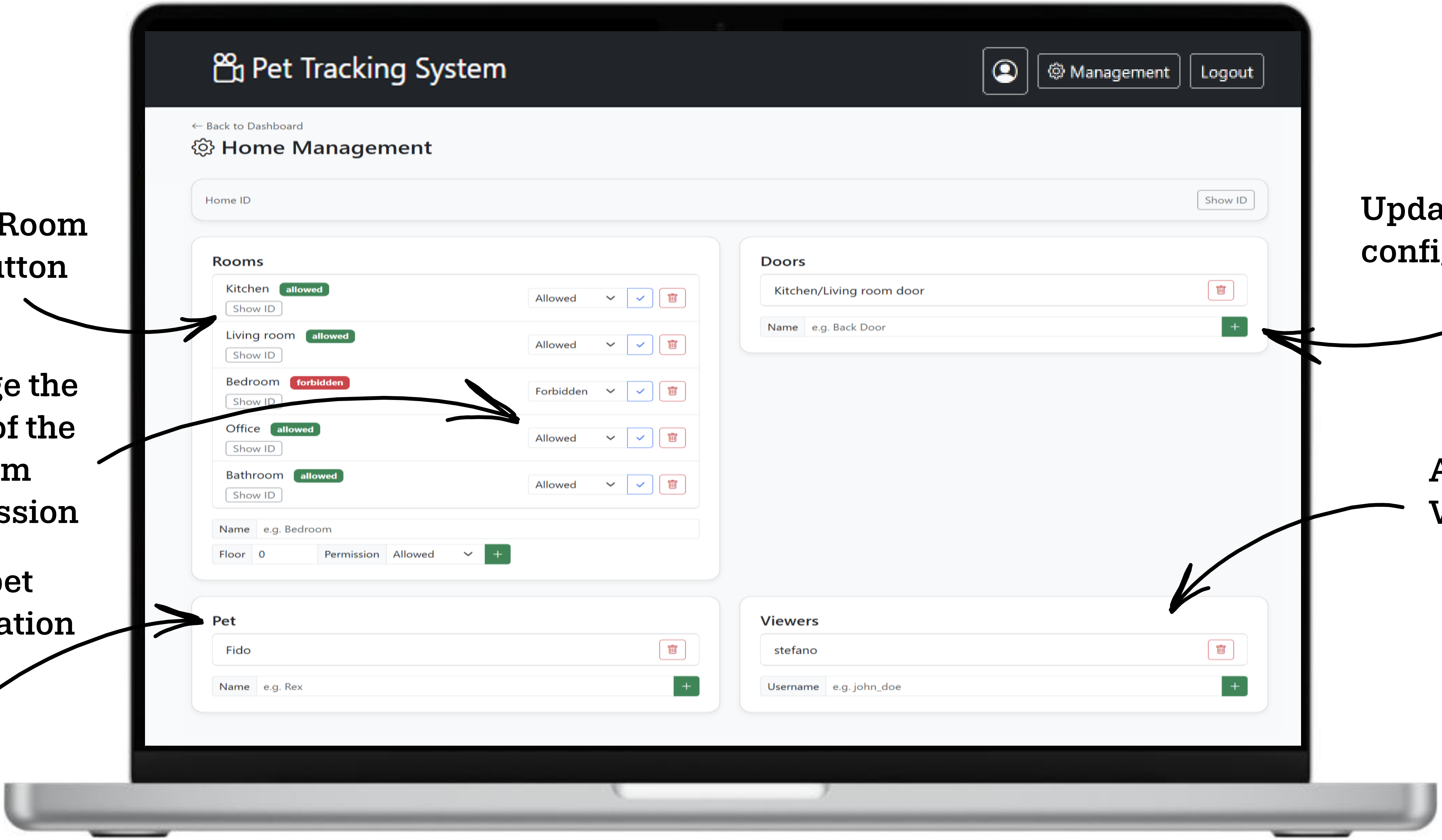
Query the system for the pet current position

Management

Show Room ID button

Change the state of the room permission

Update pet configuration



Update door configurations

Add /Remove Viewers

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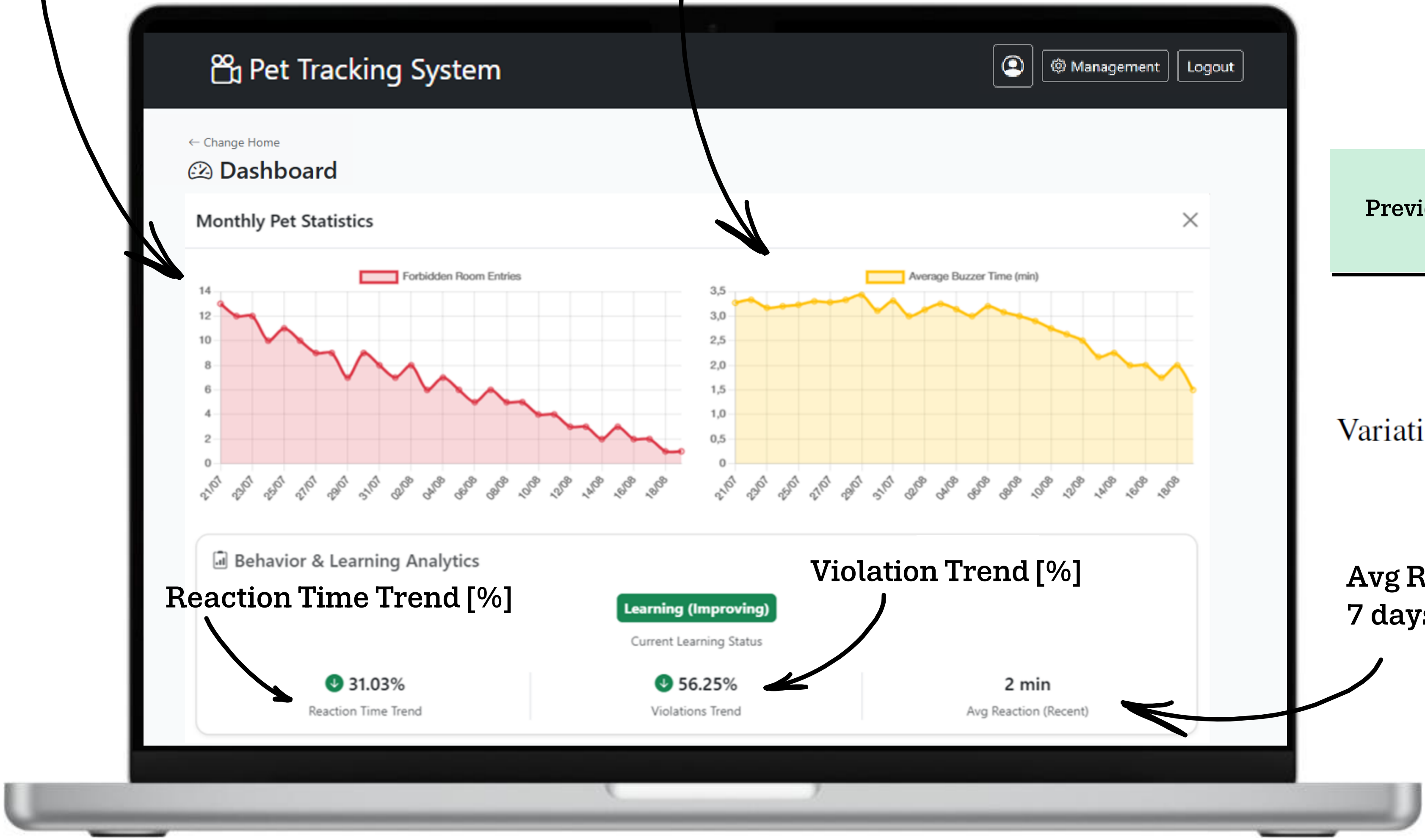
Pet Charts

Forbidden Room Entries per day

Average buzzer time [min] per day

$$\text{Daily Average Reaction Time} = \frac{\text{Daily Buzzer Duration (mins)}}{\text{Daily Violations Count}}$$

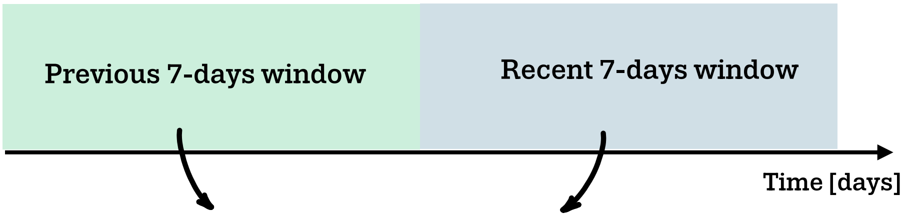


Reaction Time Trend [%]

Violation Trend [%]

Avg Reaction (Recent 7 days windows) [min]

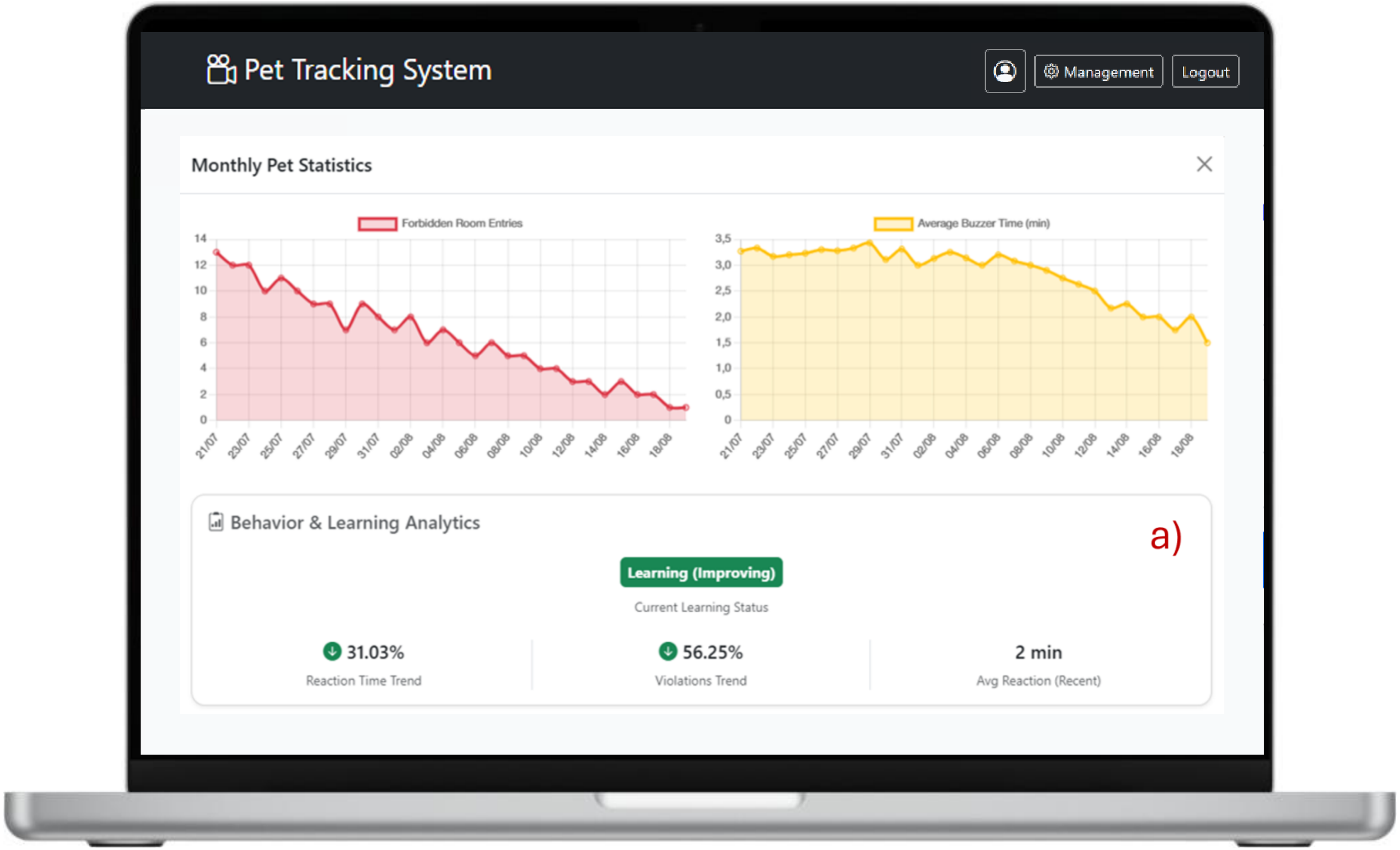
DATA ANALISYS



$$\text{Variation \%} = \left(\frac{\text{Recent Value} - \text{Previous Value}}{\text{Previous Value}} \right) \times 100$$

Pet Charts

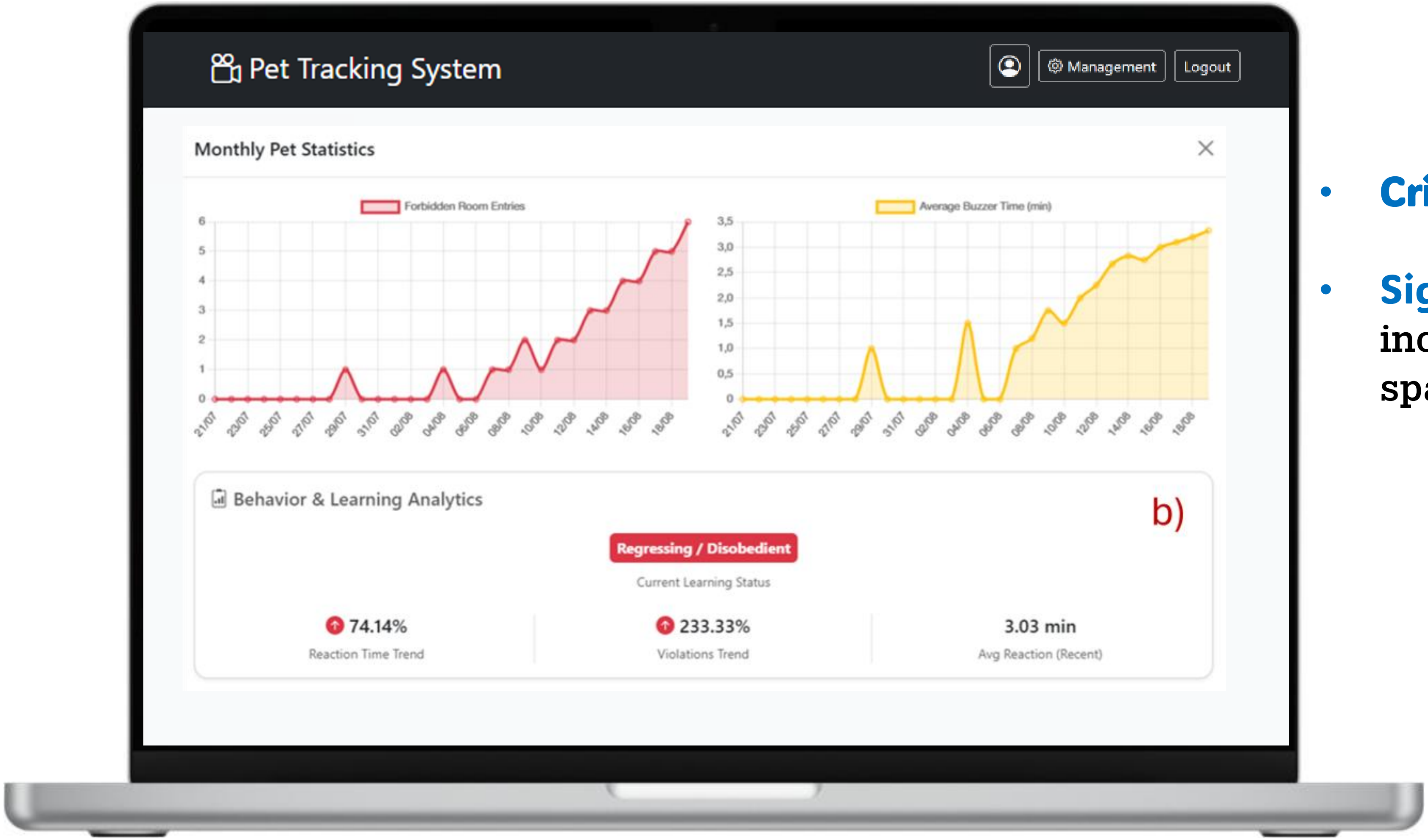
Learning (Improving)



Criterion: Variation $< -10\%$

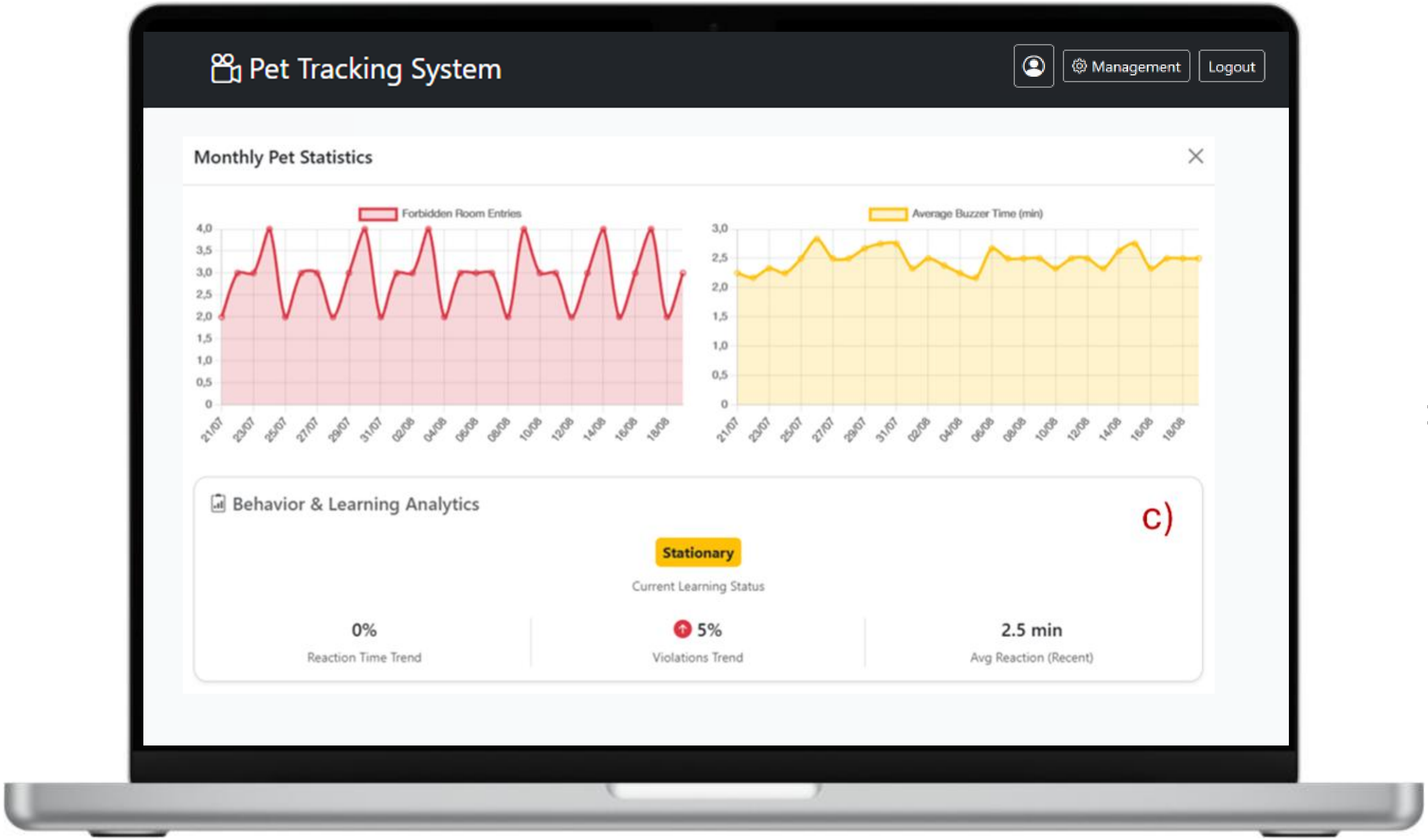
Significance: Positive response to deterrent; clear understanding of boundaries.

Regressing/Disobedient



- **Criterion**: Variation > 10%
- **Significance**: Performance decline; increased tendency to enter forbidden spaces.

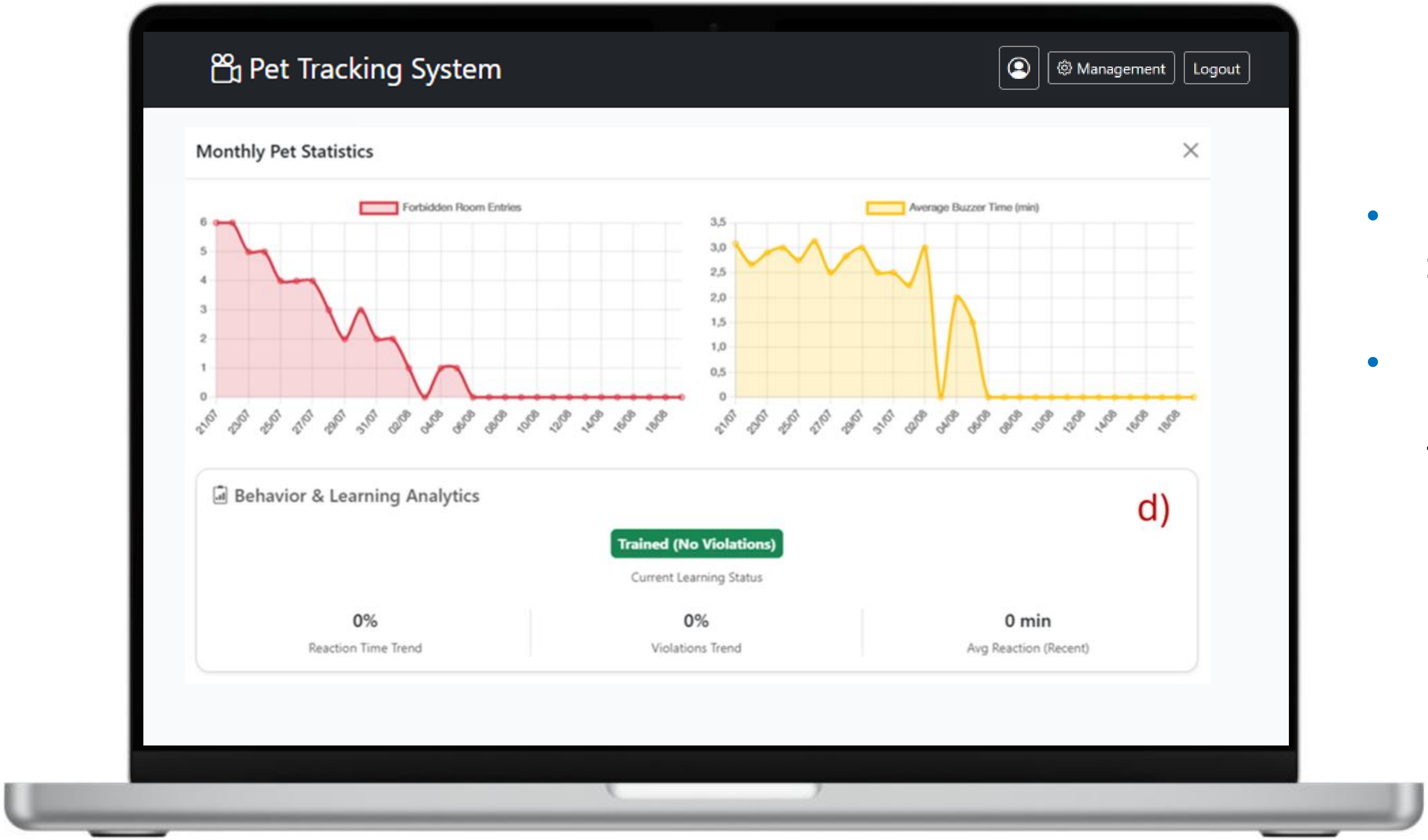
Stationary



Criterion: Variation **within** $[-10\%, +10\%]$ **threshold.**

Significance: Stable performance; no significant improvement or deterioration.

Trained (no Violations)



- **Criterion**: Zero violations in the recent window.
- **Significance**: Successful training; consistent respect for restricted boundaries.

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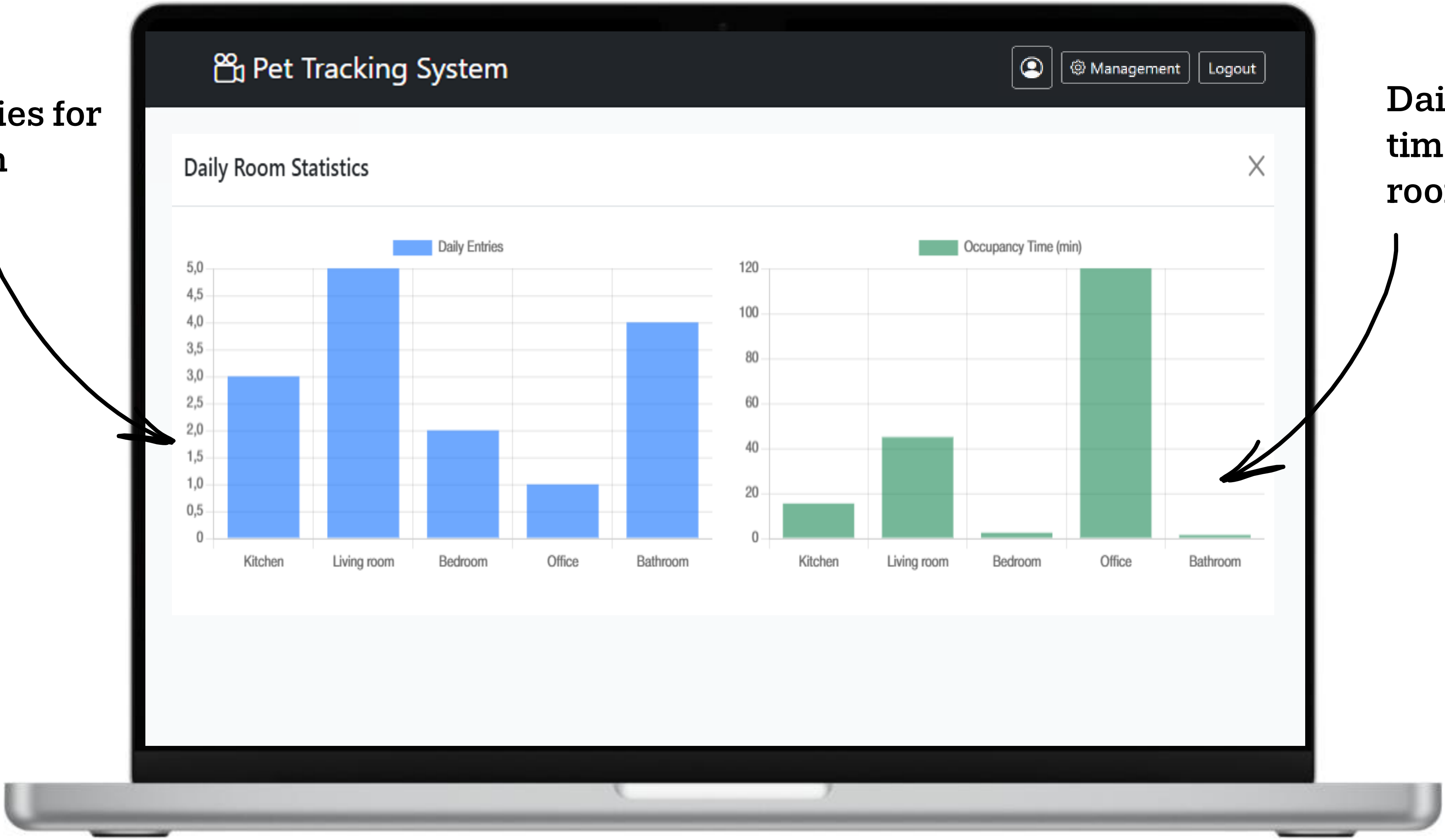
Room Charts

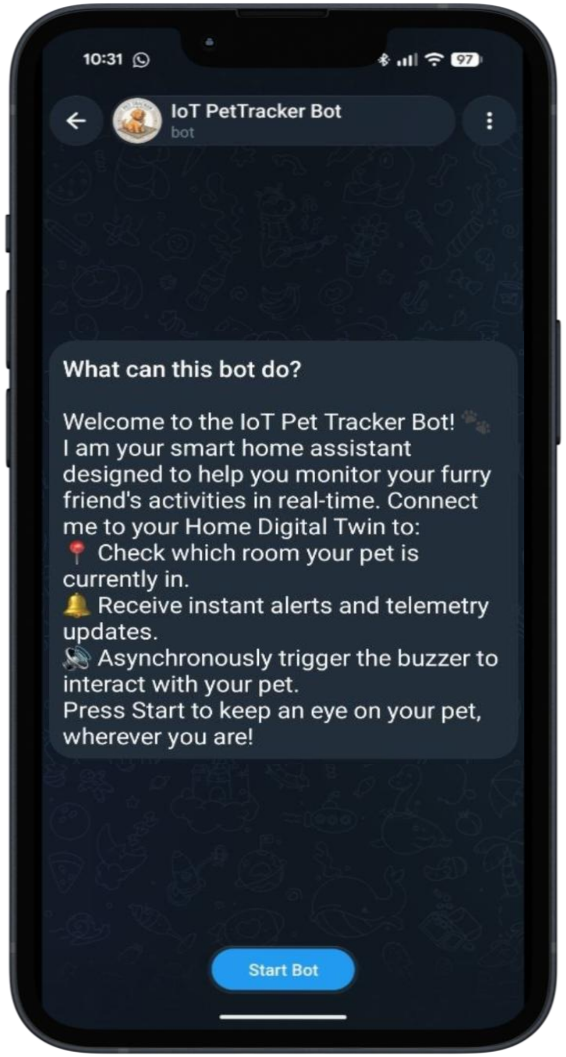
Daily entries for each room



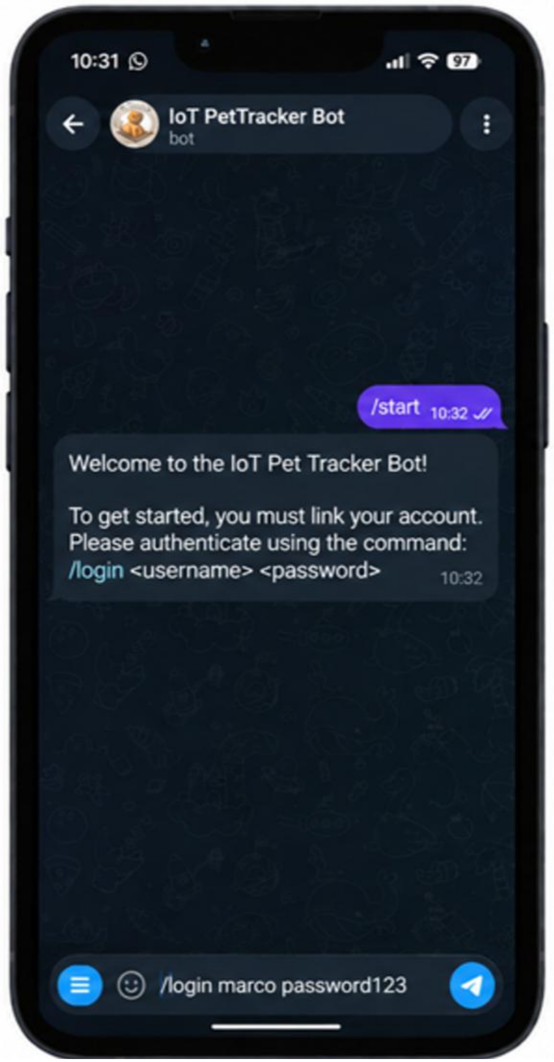
Daily occupancy time for each room [min]

The Telegram Bot- Start Bot and login

Main informations

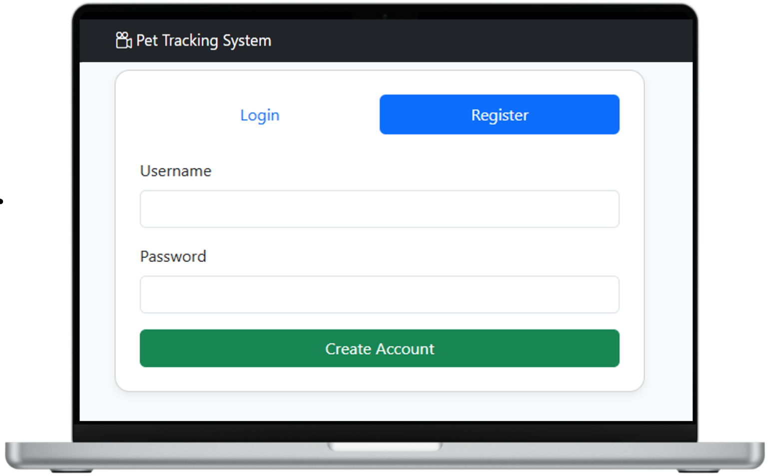


2° step - **/login**



next

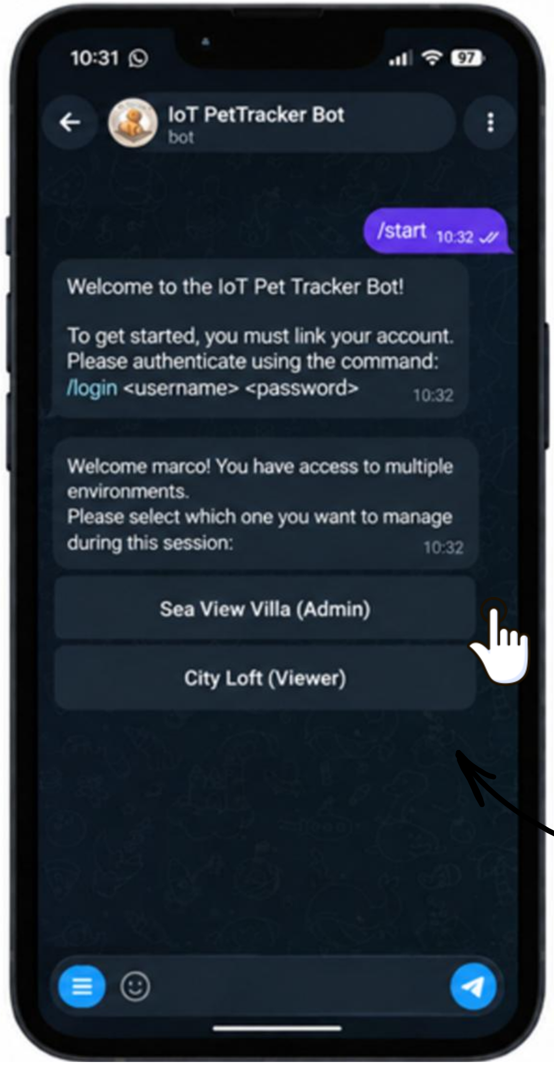
1° step - **Registration**



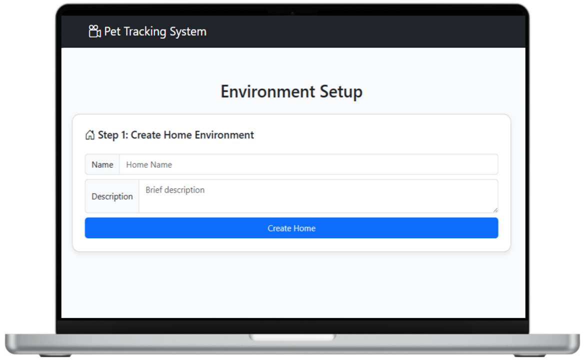
Account registration on the Web App is required before logging into the Telegram Bot (use the same credentials).

The Telegram Bot- Home Selection and logout

2° step - **Home selection**

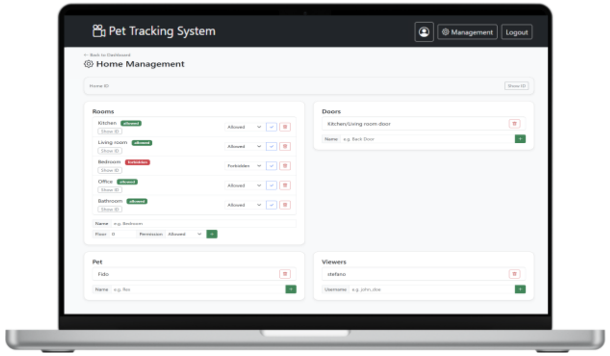


1° step – **Create an Home**



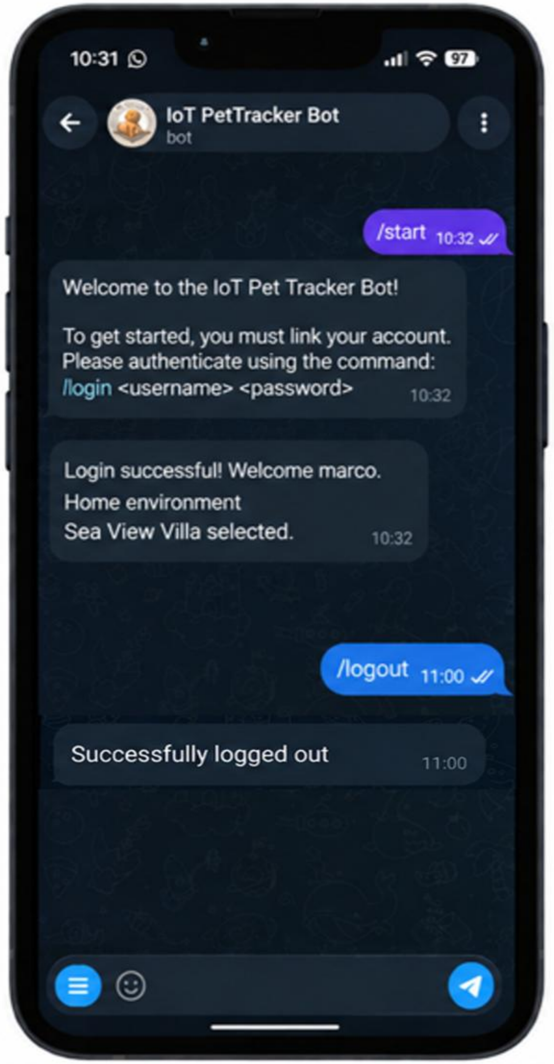
OR

Be added as Viewer



Select an accessible environment based on user role (Admin or Viewer).

/logout



Users can explicitly end their session at any time

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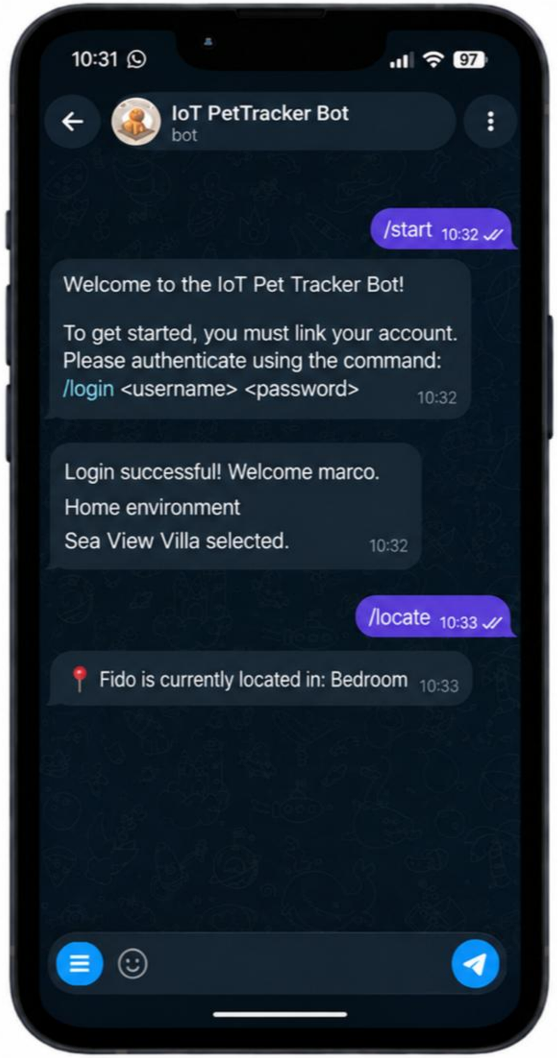
The Telegram Bot- \locate and \buzzer commands

/locate



Query the system for real-time pet position

/buzzer

Only the administrator can activate/deactivate the buzzer when the pet is in an allowed room

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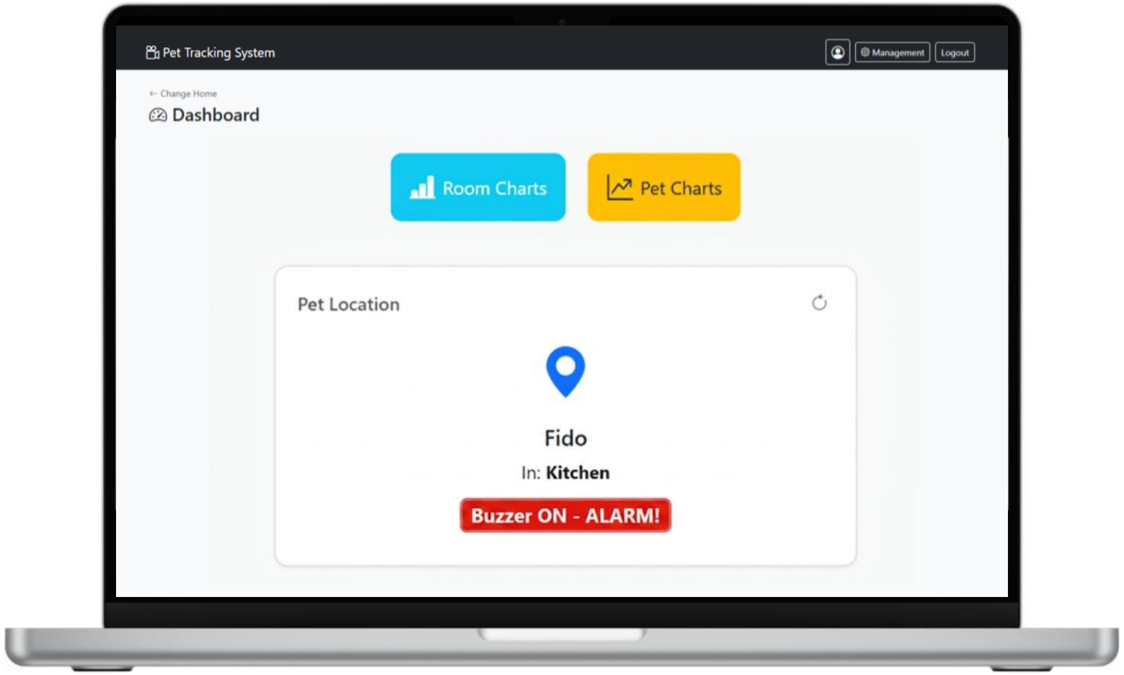
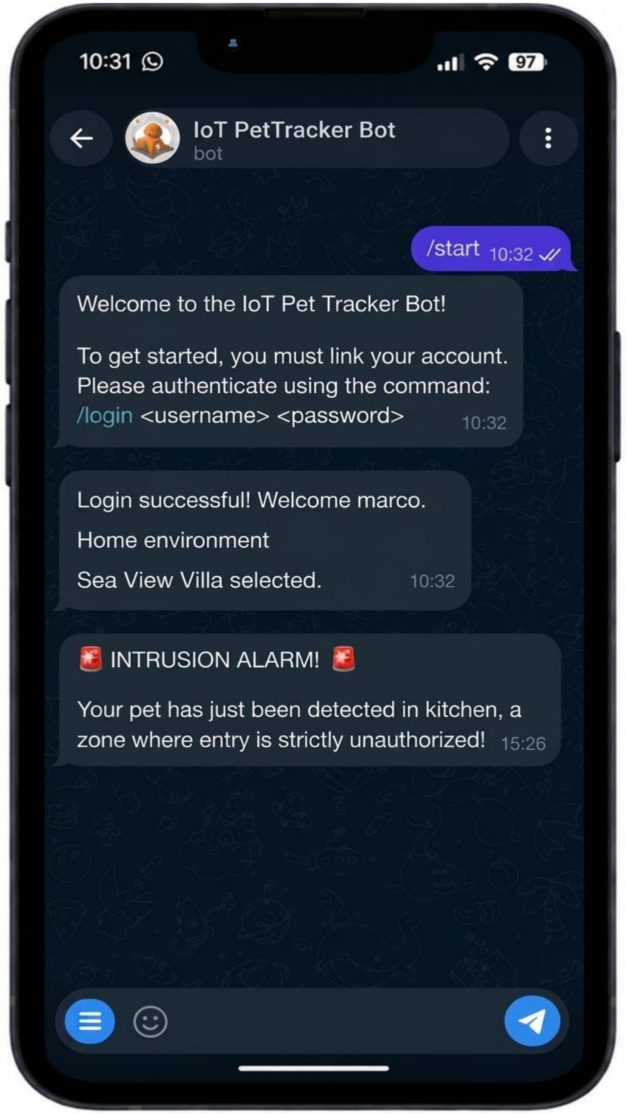
The Telegram Bot - Intrusion into a forbidden room



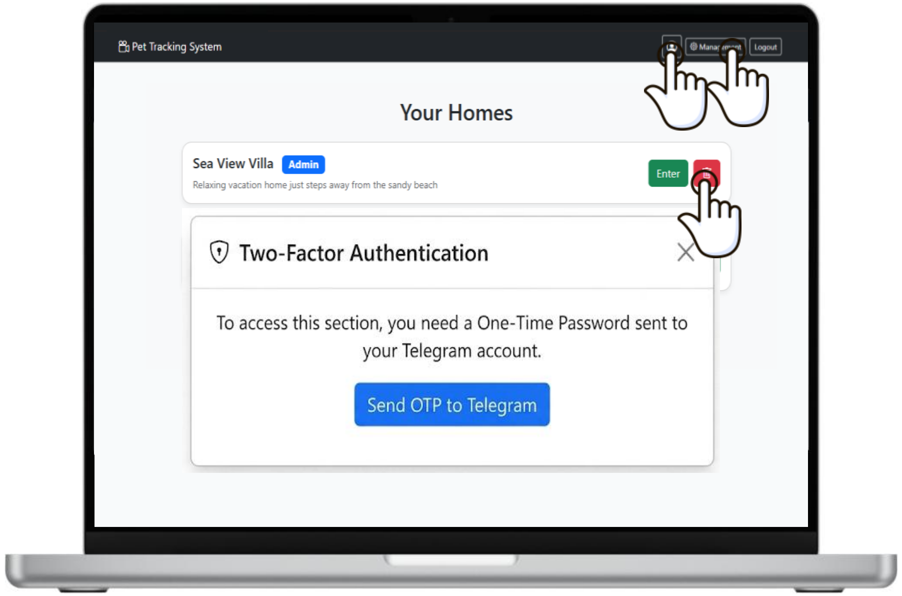
When the pet enters a forbidden room, the system automatically pushes a notification to all the logged users.

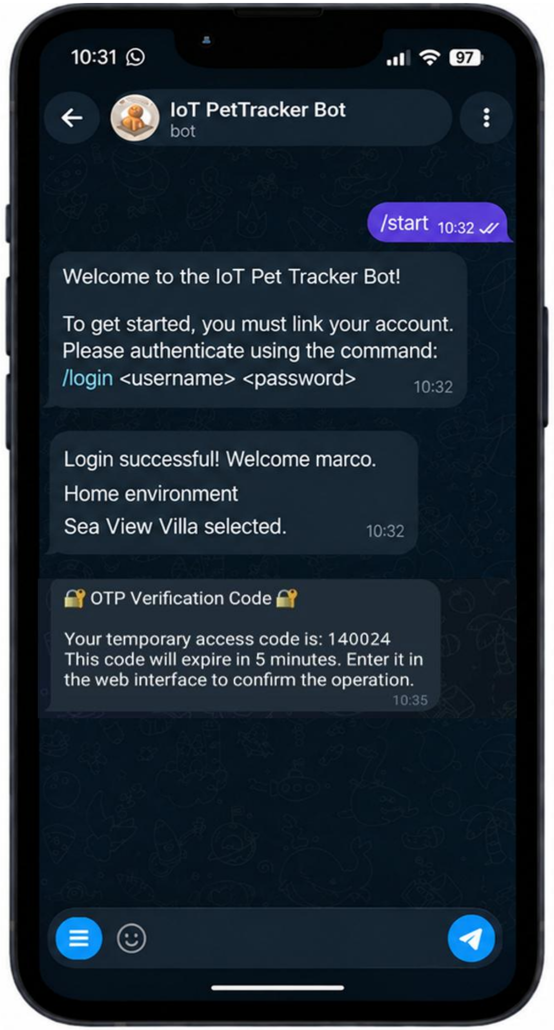


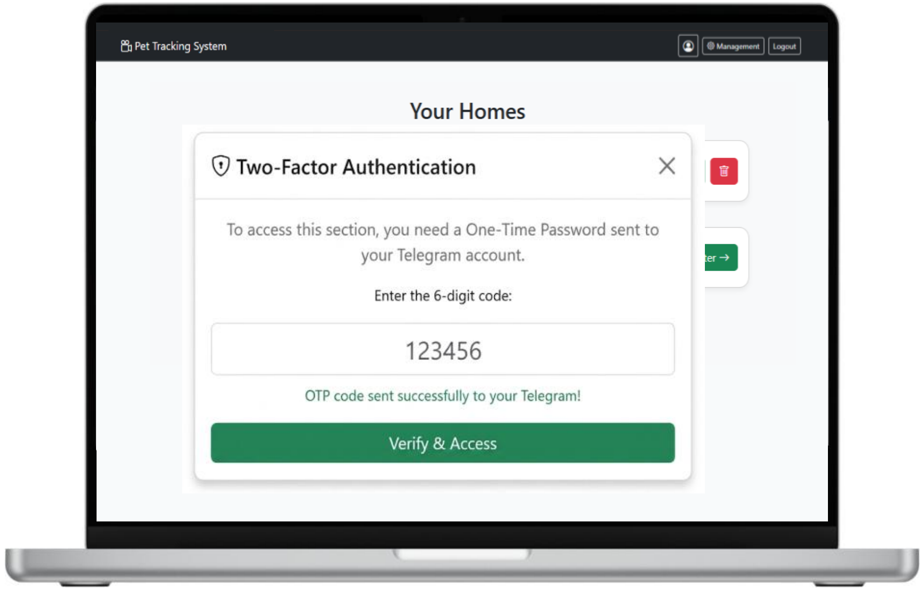
The Telegram Bot – Critical Operations

CRITICAL OPERATIONS

- Delete the Home Environment
- Access to the Management section
- Acess to the User Profile







After pressing the "Send OTP to Telegram" button, the OTP verification code is sent to the bot

The 6-digit code has to be inserted here

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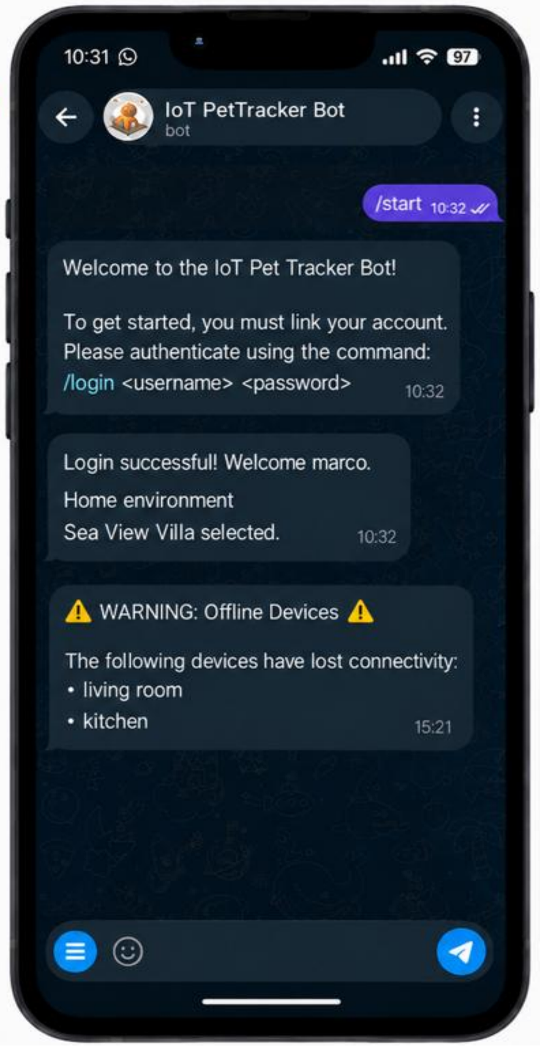
The Telegram Bot- State System Notifications

Users know the system's connection status in real time.

A



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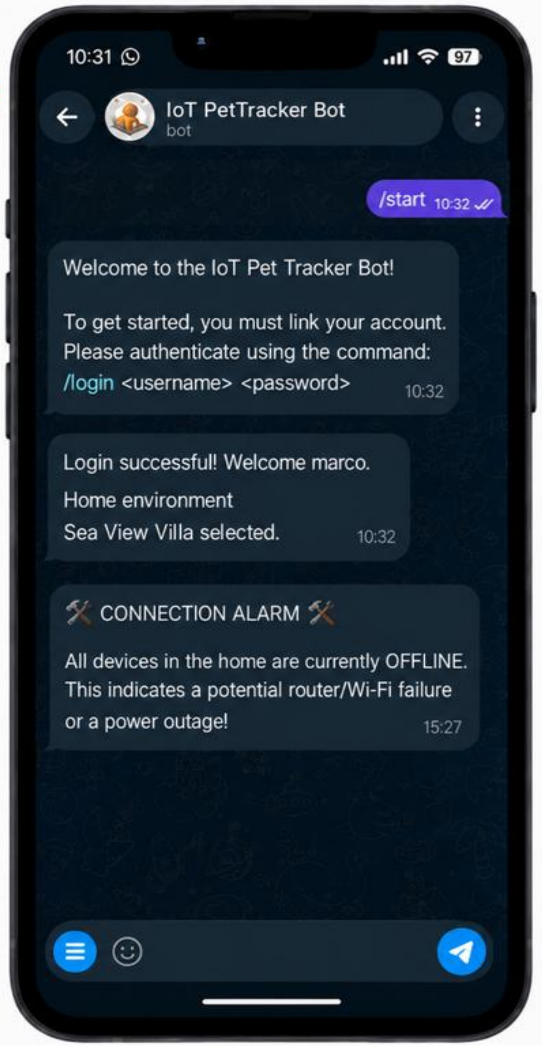
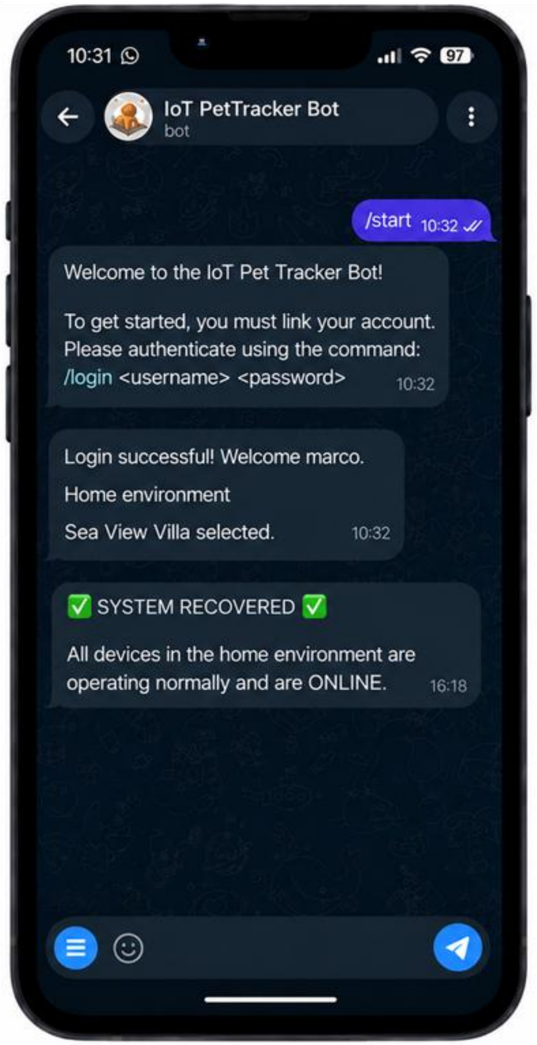


C



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Pet-Tracker

Thanks for the attention!

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